Experimento:

10B, 15N

Una urna contiene 10 bolillas blancas y 5 negras. Se tira una moneda al aire. Si sale cara se agregan 10 bolillas negras a la urna; si sale ceca se agregan 10 bolillas blancas. Se saca una bolilla al azar y se observa el color.

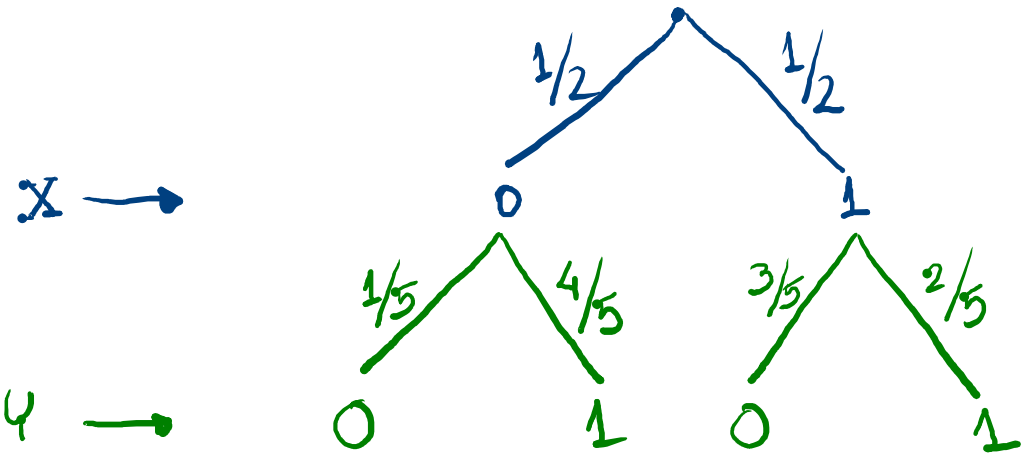
↳ 20B, 5N

Se definen dos variables aleatorias discretas:

X : vale 0 si sale ceca al arrojar la moneda y 1 si sale cara

Y : vale 0 si la bolilla extraída es negra y 1 si es blanca.

Y \ X	0	1	P_Y
0	0.1	0.3	0.4
1	0.4	0.2	0.6
P_X	0.5	0.5	1 \ 1



$$\underbrace{P(X=0, Y=0)}_{\text{función de masa de probabilidad conjunto}} = P_{X,Y}(0,0) = P(Y=0|X=0)\cdot P(X=0) = \underbrace{P_{Y|X}(0|0)}_{\text{f.m.p. condicional}}\cdot \underbrace{P_X(0)}_{\text{f.m.p. marginal}}$$

$$\left.\begin{aligned} P_{X,Y}(0,0) &= 0.1 \\ P_{X,Y}(0,1) &= 0.4 \\ P_{X,Y}(1,0) &= 0.3 \\ P_{X,Y}(1,1) &= 0.2 \end{aligned}\right\} \boxed{\sum_{\beta \in R_Y} \sum_{\alpha \in R_X} P_{X,Y}(\alpha, \beta) = 1}$$

$$\left.\begin{aligned} P_X(0) &= P_{X,Y}(0,0) + P_{X,Y}(0,1) \\ P_X(1) &= P_{X,Y}(1,0) + P_{X,Y}(1,1) \end{aligned}\right\} \boxed{P_X(\alpha) = \sum_{\beta \in R_Y} P_{X,Y}(\alpha, \beta)}$$

$$\left.\begin{aligned} P_Y(0) &= P_{X,Y}(0,0) + P_{X,Y}(1,0) \\ P_Y(1) &= P_{X,Y}(0,1) + P_{X,Y}(1,1 \end{aligned}\right\} \boxed{P_Y(\beta) = \sum_{\alpha \in R_X} P_{X,Y}(\alpha, \beta)}$$

$$\boxed{E[g(X,Y)] = \sum_{\alpha \in R_X} \sum_{\beta \in R_Y} g(\alpha,\beta) \cdot P_{X,Y}(\alpha,\beta)}$$

Valor esperado

$$E[h(X)] = \sum_{\alpha \in R_X} \sum_{\beta \in R_Y} h(\alpha) \cdot P_{X,Y}(\alpha,\beta) = \sum_{\alpha \in R_X} h(\alpha) \sum_{\beta \in R_Y} P_{X,Y}(\alpha,\beta)$$

$$= \sum_{\alpha \in R_X} h(\alpha) P_X(\alpha) = E[h(X)]$$

Algo similar para funciones sólo de Y